

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T12 · Ratio Toolkit

With Answers

Questions with full worked solutions

■ 5 pages

FAMILY MAP

Ratio Toolkit

F01

Combine linked ratios

2 Q | 7 marks

Question 25

4MA1-2H Q5 whole

ORIGINAL QUESTION UNIT 2

3 marks

5 Avril bakes a cake.

She uses flour, butter and sugar such that

$$\text{weight of flour : weight of butter} = 6 : 5$$
$$\text{weight of butter : weight of sugar} = 3 : 2$$

Avril uses 120 grams of sugar.

Work out the weight of flour Avril uses.

..... grams

Worked solution 25

Chained ratio of flour, butter and sugar

MODULAR UNIT 2

3 marks

whole – Chained ratio of flour, butter and sugar

Find the butter amount

$$\text{Butter:sugar} = 3 : 2, \quad 120 \div 2 = 60, \quad \text{butter} = 3(60) = 180 \text{ g.}$$

Use the flour:butter ratio

$$\frac{\text{flour}}{180} = \frac{6}{5} \Rightarrow \text{flour} = \frac{6}{5}(180) = 216 \text{ g.}$$

Final answer

216 grams

Question 19

4MA1-2H Q5

ORIGINAL QUESTION

4 marks

- 5 Nancy has some coins with a total value of 85 pence.

She has only 2 pence coins and 5 pence coins.

The ratio

$$\text{number of 2 pence coins} : \text{number of 5 pence coins} = 1 : 3$$

Nancy has more 5 pence coins than 2 pence coins.

How many more?

Worked Solution 19

Value-weighted coin ratio

MODULAR UNIT 2

4 marks

Write the value ratio

1 : 3 coins gives value ratio 2 : 15.

Use the total

$$2n + 15n = 85 \Rightarrow 17n = 85 \Rightarrow n = 5.$$

Count and subtract

There are 5 two-pence coins and 15 five-pence coins, so $15 - 5 = 10$.

Final answer

10